

1. Introduction

1.1 Purpose

This document describes the software requirements for the *Invisible Cloak Project*. The system is designed to create a real-time invisibility effect by detecting a specific color and replacing it with a captured background using computer vision techniques.

1.2 Document Conventions

- Headings and subheadings are used to organize content.
- Requirements are labeled as REQ-1, REQ-2, etc.
- Technical terms such as HSV, masking, and segmentation are used consistently.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Students and developers implementing the project
- Faculty evaluating the project
- Beginners learning computer vision

Readers should start with the introduction and proceed to system features and requirements.

1.4 Product Scope

The system captures a video stream, detects a predefined color (pink), and replaces it with a stored background to create an invisibility effect. The project demonstrates key concepts in computer vision such as color detection, masking, and real-time processing.

2. Overall Description

2.1 Product Perspective

This is a standalone system developed for educational purposes. It uses a webcam as input and processes video frames using Python and OpenCV.

2.2 Product Functions

- Capture background image
- Detect specific color in real-time
- Generate mask for selected color
- Replace masked region with background
- Display processed output

2.3 User Classes and Characteristics

- **Students/Developers:** Basic knowledge of Python
- **Evaluators:** Interested in functionality and output
- **Beginners:** Learning computer vision concepts

2.4 Operating Environment

- Operating System: Windows / Linux
- Programming Language: Python
- Libraries: OpenCV, NumPy
- Hardware: Webcam

2.5 Design and Implementation Constraints

- Works best with controlled lighting
- Requires a static background
- Detects only one predefined color
- Limited by webcam quality

2.6 User Documentation

- Project report
- Source code comments
- Basic usage instructions

2.7 Assumptions and Dependencies

- User provides proper lighting conditions
- Required libraries are installed
- Camera is functional

3. External Interface Requirements

3.1 User Interfaces

- Output is displayed using OpenCV window
- Simple keyboard control (ESC to exit)

3.2 Hardware Interfaces

- Webcam for capturing video input

3.3 Software Interfaces

- Python interpreter

- OpenCV library
- NumPy library

3.4 Communications Interfaces

Not applicable (no network communication required)

4. System Features

4.1 Background Capture

4.1.1 Description and Priority

Captures	static	background	before	processing
Priority: High				

4.1.2 Stimulus/Response

- User starts program → System captures background

4.1.3 Functional Requirements

REQ-1: System shall capture background frames

REQ-2: System shall store background for later use

4.2 Color Detection and Masking

4.2.1 Description and Priority

Detects predefined color using HSV
Priority: High

4.2.2 Stimulus/Response

- Color detected → Mask created

4.2.3 Functional Requirements

- REQ-3: System shall convert frame to HSV
- REQ-4: System shall detect predefined color range
- REQ-5: System shall generate mask

4.3 Invisibility Effect

4.3.1 Description and Priority

Replaces detected region with background

Priority: High

4.3.2 Stimulus/Response

- Mask applied → Cloak disappears

4.3.3 Functional Requirements

- REQ-6: System shall replace masked region
- REQ-7: System shall display final output

5. Nonfunctional Requirements

5.1 Performance Requirements

- System should process video in real-time
- Minimal delay between input and output

5.2 Safety Requirements

- No physical risk to users
- Safe usage of webcam

5.3 Security Requirements

- No sensitive data stored
- No external access required

5.4 Software Quality Attributes

- **Usability:** Easy to run and understand
- **Reliability:** Works under proper conditions
- **Maintainability:** Simple code structure
- **Portability:** Can run on different OS

6. Project Plan

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6.1 Division of Work

- Requirement analysis
- Design and coding
- Testing and debugging

Appendix A: Glossary

- **HSV:** Color model used for detection
- **Masking:** Identifying specific image regions
- **OpenCV:** Computer vision library
- **Frame:** Single image from video

Appendix B: To Be Determined List

- Future enhancements (GUI, multi-color detection)